Name:_____

1 Piecewise linear approximations

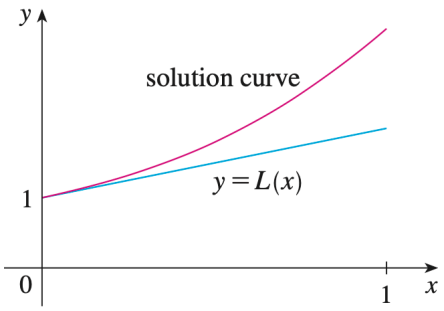
The basic idea behind direction fields can be used to find numerical approximations to solutions of differential equations. To illustrate this method, consider the initial-value problem

$$y' = x + y, \quad y(0) = 1.$$

The differential equation tells us that $y'(0) = 0 + 1 = 1$, so the solution curve has slope 1 at the point $(0, 1)$. As a first approximation to the solution we could use the linear approximation

$$L(x) = x + 1. \tag{1.1}$$

In other words, we could use the tangent line at $(0, 1)$ as a rough approximation to the solution curve, as shown below:



Euler's idea was to improve on this approximation by proceeding only a short distance along the tangent line and then making a midcourse correction by changing direction as indicated by the direction field.

1. Suppose we stop at $x = 0.5$. Use the linear approximation in Equation (1.1) to approximate $y(0.5)$.

2. Find $y'(0.5)$.

3. Find the equation of the tangent line to the graph at $(0.5, y(0.5))$.

4. Use the linear approximation in question 3 to approximate $y(1)$.

5. Graph the piecewise-linear function made up of these two linear approximations (Equation (1.1) and question 3) on the graph above. Does this give a better approximation to the solution on the interval $[0, 1]$?

6. Explain how you could continue this process to approximate the solution on the interval $[0,2]$.

7. How do you think you can get a better approximation?

2 Euler's Method

We now formalize the process above into an algorithm, which is due to Swiss mathematician Leonard Euler (1707-1783):

Consider a first-order initial value problem

$$y' = F(x,y), \quad y(x_0) = y_0.$$

Our aim is to find approximate values for the solution at equally-space numbers $x_0, x_1 = x_0 + \Delta x$, $x_2 = x_1 + \Delta x$, ..., where Δx is called the *step-size*.

(1) The initial value says that the solution passes through (x_0, y_0).

(2) The differential equation says the slope at (x_0, y_0) is $F(x_0, y_0)$, so the linear approximation is $y(x) = y_0 + F(x_0, y_0)(x - x_0)$. The approximate solution at $x = x_1$ is therefore

$$y_1 = y_0 + F(x_0, y_0)\Delta x.$$

(3) The linear approximation to the curve at (x_1, y_1) is then $L(x) = y_1 + F(x_1, y_1)(x - x_1)$, so the approximate solution at $x = x_2$ is therefore

$$y_2 = y_1 + F(x_1, y_1)\Delta x.$$

(4) Continuing in this way, the approximate solution at $x = x_n$, where $n \geq 1$ is

$$\boxed{y_n = y_{n-1} + F(x_{n-1}, y_{n-1})\Delta x}.$$

In other words, in each step, after finding the old solution y_{old}, the new solution y_{new} is given by $y_{\text{new}} = y_{\text{old}} + F(x_{\text{old}}, y_{\text{old}})\Delta x$.

3 Practice

8. Use Euler's method with step size 0.1 to construct a table of approximate values for the solution to the initial-value problem

$$y' = x + y, \quad y(0) = 1.$$

9. We previously discussed a simple electric circuit with resistance 12 Ω, inductance $4H$, and a battery with voltage $60V$. If the switch is closed when $t = 0$, we modeled the current I at time t by the initial-value problem

$$\frac{dI}{dt} = 15 - 3I, \quad I(0) = 0.$$

Use Euler's method with step-size 0.1 to estimate the current in the circuit half a second after the switch is closed.